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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-34 20kV FlourMill Panel Type : UNISEC Customer Name : PT WILMAR NABATI INDONESIA Contract No. : BE318043 TITLE : SLD & GA Drawing								C																																												
D									D																																												
E									E																																												
F	<table><tr><td>2</td><td>Issued for Approval</td><td>12.02.2019</td><td>YR</td><td>Date Prep.</td><td>15.11.2018</td><td rowspan="3">PROJECT TITLE</td><td rowspan="3"> ABB PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>SLD & GA Drawing</td><td>Customer</td><td>PT WILMAR NABATI INDONESIA</td><td>+</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td>MVD-34 20kV FlourMill</td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318043</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>								2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE	ABB PT. ABB Sakti Industri	COVER SHEET	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	SLD & GA Drawing	Customer	PT WILMAR NABATI INDONESIA	+	0	First Issued	19.11.2018	YR	Checked	AW	MVD-34 20kV FlourMill	Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043		Customer Doc. Nbr.		Cont. 16	F
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A	SHEET INDEX								A							
B	PAGE		PAGE NAME			REVISION			B							
	==MVD-34=H00&GAD/1		COVER SHEET			0	1	2								
	==MVD-34=H00&GAD/2		INDEX OF SHEETS			0	1	2								
	==MVD-34=H00&GAD/3		GENERAL CHARACTERISTICS			0	1	2								
	==MVD-34=H00&GAD/4		REFERENCE DESIGNATIONS			0	1	2								
	==MVD-34=H00&GAD/5		REFERENCE DESIGNATIONS			0	1	2								
	==MVD-34=H00&GAD/6		REFERENCE DESIGNATIONS			0	1	2								
	==MVD-34=H00&GAD/7		REFERENCE DESIGNATIONS			0	1	2								
	==MVD-34=H00&GAD/8		SINGLE LINE DIAGRAM			0	1	2								
	==MVD-34=H00&GAD/9		INSTALLATION RULES			0	1	2								
	==MVD-34=H00&GAD/10		FRONT VIEW & GA			0	1	2								
	==MVD-34=H00&GAD/11		FOUNDATION FRAMES			0	1	2								
	==MVD-34=H00&GAD/12		FOUNDATION FRAMES DETAILS			0	1	2								
	==MVD-34=H00&GAD/13		FOUNDATION FRAMES DETAILS			0	1	2								
	==MVD-34=H00&GAD/14		SECTION VIEW			0	1	2								
	==MVD-34=H00&GAD/15		SECTION VIEW			0	1	2								
C	==MVD-34=H00&GAD/16		SECTION VIEW			0	1	2	C							
D									D							
E									E							
F	2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-34 20kV FlourMill		Scale		= H00	F
	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant						Customer	PT WILMAR NABATI INDONESIA & GAD		
	0	First Issued	19.11.2018	YR	Checked	AW							PT. ABB Sakti Industri		Drawing Nbr.	
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A												
B	SWITCHGEAR VERSION			= UNISEC	COLOUR OF FRONT DOORS			= RAL 7035	B												
	RATED VOLTAGE (Un)			= 24 kV	SIDEWALLS PAINTED			= YES													
	OPERATING VOLTAGE			= 20 kV	SIDES			= L+R													
	RATED FREQUENCY (fr)			= 50 Hz																	
C	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)			= 125 kV	SUPPLIED IMPLEMENTS					C											
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)			= 50 kV																	
	RATED CURRENT OF MAIN BUSBARS (Ir)			= 630 A																	
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)			= 20 kA																	
D	RATED PEAK WITHSTAND CURRENT (Ip)			= 50 kA	IDENTIFICATION OF CONDUCTORS					D											
	RATED DURATION OF SHORT CIRCUIT (tk)			= 1 s																	
	INTERNAL ARC CLASSIFICATION (IAC)			= AFLR / 16kAx1s																	
	AMBIENT CONDITION			= NORMAL																	
E	AMBIENT AIR TEMPERATURE			= -5°C...+40°C						FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW -XDA OT:VdX- 91011 -XDA:10					E						
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)			= IP3X																	
	DEGREE OF PROTECTION WITH OPEN DOORS			= IP2X																	
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)			= 230VAC																	
F	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)			= 230VAC											F						
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)			= 230VAC																	
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= 450/750V																	
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= PVC																	
	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS			= 2.5 mm²/BLACK																	
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS			= 1.5 mm²/BLACK																	
	CROSS-SECTION OF CONDUCTORS FOR MOTORS			= 2.5 mm²/BLACK																	
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)			= 1.5 mm²/BLACK																	
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS			= 1.5 mm²/BLACK																	
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE			= 2.5 mm²/BLACK																	
	COMMUNICATION CABLE			= N/A																	
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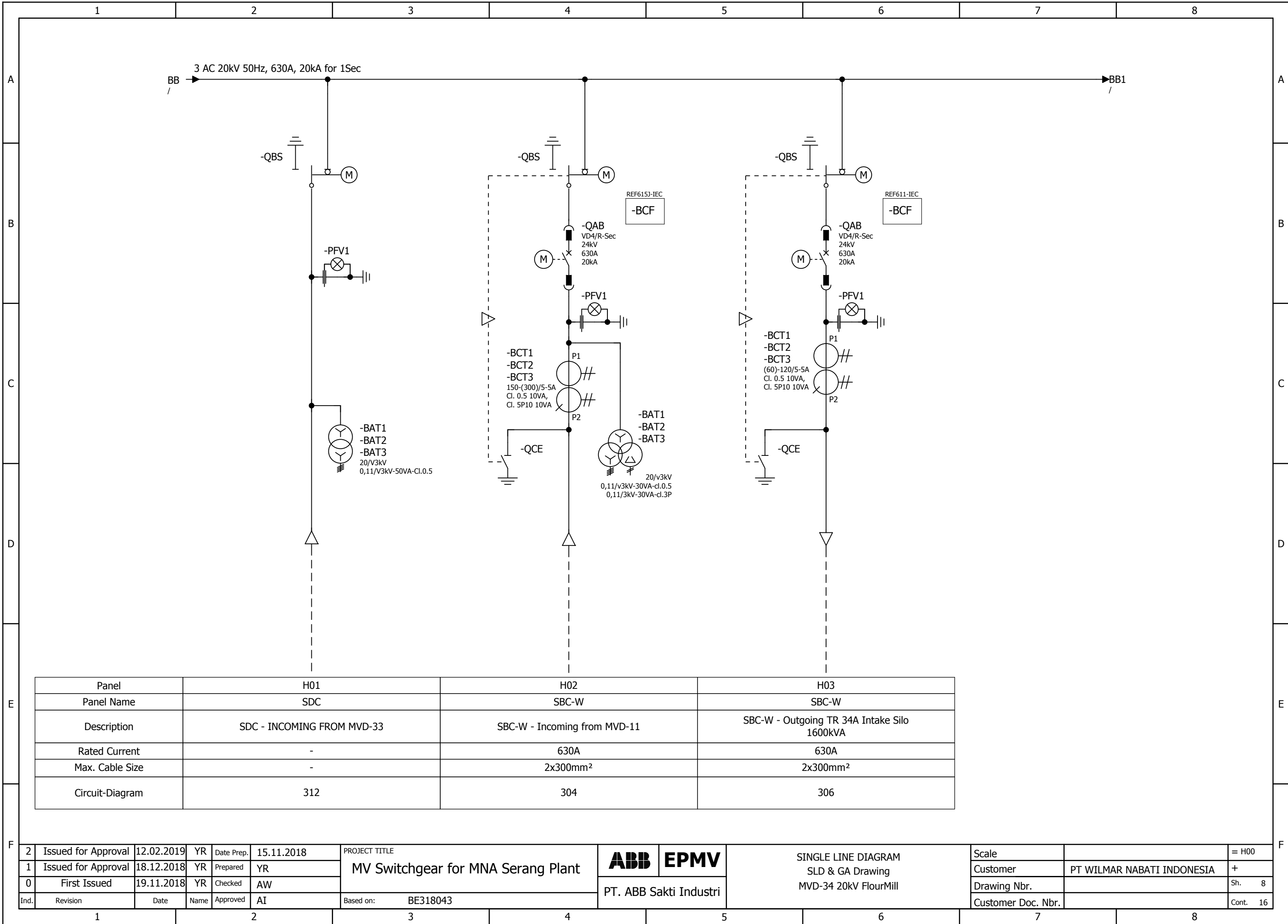
A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION										-CA	CAPACITORS										A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION										-CA1	CAPACITOR FOR FRONT VENTILATOR										
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)										-CA2	CAPACITOR FOR REAR VENTILATOR										
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION										-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.										
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION										-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT										
	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION										-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE										
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION										-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT										
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION										-EB1	HEATER LOCATED IN CABLE COMPARTMENT										
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION										-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT										
	-BGK		POSITION SWITCH OF KEY LOCK										-EB3	HEATER LOCATED IN MOTOR CABINET										
B	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1										-EB4	HEATER FOR VENTILATION TEST										B
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS										-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.										
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION										-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION										-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE										
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION										-FA1	SURGE ARRESTER LOCATED ON PHASE L1										
C	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION										-FA2	SURGE ARRESTER LOCATED ON PHASE L2										C
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION										-FA3	SURGE ARRESTER LOCATED ON PHASE L3										
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION										-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION										-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1										
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION										-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2										
	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION										-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3										
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION										-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER										
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION										-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS										
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR										-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1										
	-BGV2		POSITION SWITCH OF REAR VENTILATOR										-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2										
D	-BM		HYGROSTAT										-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3										D
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-FCT	THERMAL OVERLOAD RELEASES										
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GA	GENERATORS										
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GQ1	FRONT VENTILATOR										
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING										-GQ2	REAR VENTILATOR										
E	-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER										-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK										E
	-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2										-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE										
	-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3										-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE										
	-BR		FLAME DETECTORS, SMOKE DETECTORS										-KFA3....-KFA9	AUXILIARY RELAYS OR CONTACTORS										
	-BT		THERMOSTAT										-KFC	CLOSING RELAYS OR CONTACTORS										
	-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS										-KFI	INTEGRATED CIRCUITS										
	-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS										-KFL	LOCKOUT RELAY										
	-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM										-KFM	MICROPROCESSORS										
	-BU		BUCHHOLZ RELAY										-KFN	ANTIPUMPING RELAYS										
	-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING										-KFO	OPENING RELAYS OR CONTACTORS										
F	-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING										-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)										F
	-BZ,-BZ4		MULTIFUNCTION SENSORS										-KFR	RECLOSING RELAYS										
	-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1										-KFS	SYNCHRONIZING DEVICES										
	-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2										-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS										
	-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3										-KFZ	CONTROLLERS										
-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD										-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER											
-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1										-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD											
-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2										-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1											
-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS										-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2											
												-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS											
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A	1		2		3		4		5		6		7		8		A						
	-MAE		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE										-QBH		MANUAL CIRCUIT-BREAKERS			A					
	-MAE1		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2										-QBL		LINKS								
	-MAE8		MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS										-QBM		MINIATURE SWITCH-DISCONNECTORS								
	-MAS		MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING										-QBS		SWITCH-DISCONNECTORS								
	-MAT		MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT										-QBT		TRUCK								
	-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER										-QCA		EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING								
	-MBC4		CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS										-QCE		EARTHING SWITCH								
	-MBO1		FIRST OPENING RELEASE OF CIRCUIT-BREAKER										-QCE1		EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)								
	-MBO2		SECOND OPENING RELEASE OF CIRCUIT-BREAKER										-QCE2		EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)								
	-MBO3		OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER										-QQ		CLUTCHES								
	-MBO4		OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS										-RAA		ANTIFERRORESONANCE RESISTOR								
	-MBU		UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER										-RAD		DIODES								
	-PFB		BLUE SIGNAL LAMPS										-RAI		INDUCTORS								
	-PFF		FLAG RELAYS										-RAR		RESISTORS								
	-PFG		GREEN SIGNAL LAMPS										-RF		FILTERS								
	-PFR		RED SIGNAL LAMPS										-RLE1		ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING								
	-PFS		SHORT CIRCUIT INDICATORS										-RLE2		ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT								
	-PFV		VOLTAGE INDICATORS										-RLE3		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE								
	-PFV1		VOLTAGE INDICATORS ON FEEDER SIDE										-RLE4		ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION								
	-PFV2		VOLTAGE INDICATORS ON BUSBAR SIDE										-RLE5		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD								
	-PFW		WHITE SIGNAL LAMPS										-RLE6		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1								
	-PFX		CROSS INDICATORS, ELECTROMECHANICAL INDICATORS										-RLE7		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2								
	-PFY		YELLOW SIGNAL LAMPS										-RLE8		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2								
	-PGA		AMMETERS										-RLE9		ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS								
	-PGC		COUNTERS										-RLE11		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4								
	-PGF		FREQUENCYMETERS										-RLE12		ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5								
	-PGH		HOURMETERS										-SFA		AMMETRIC SWITCHES								
	-PGI		PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE										-SFC		CONTROL SWITCHES, CLOSING PUSH-BUTTONS								
	-PGI1		HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)										-SFL		LOCKING CONTACTS								
	-PGI2		HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)										-SFO		OPENING PUSH-BUTTONS								
	-PGJ		ACTIVE ENERGY METERS										-SFR		RESET PUSH-BUTTONS								
	-PGK		REACTIVE ENERGY METERS										-SFS		SELECTOR SWITCHES								
	-PGM		MULTIFUNCTION INDICATORS										-SFT		TEST PUSH-BUTTONS								
	-PGP		POWER-FACTOR METERS										-SFU		UNLOCKING PUSH-BUTTONS								
	-PGQ		VARMETERS										-SFV		VOLTMETRIC SWITCHES								
	-PGS		SYNCHRONOSCOPES										-TA		POWER TRANSFORMERS								
	-PGV		VOLTMETERS										-TB		CONVERTER								
	-PGW		WATTMETERS										-TB1		RECTIFIER FOR FIRST OPENING RELEASE								
	-PJ		ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)										-TB2		RECTIFIER FOR SECOND OPENING RELEASE								
	-QAB		CIRCUIT-BREAKERS										-TB3		RECTIFIER FOR CLOSING RELEASE								
	-QAC		CONTACTORS (FOR POWER)										-TB4		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING								
	-QBD		DISCONNECTORS										-TB5		RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT								
	-QBD1		FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)										-TB6		RECTIFIER FOR UNDERVOLTAGE RELEASE								
	-QBD2		REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)										-TB7		ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER								
	-QBD3		FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)																				
	-QBD4		TWO-WAY DISCONNECTOR																				
	-QBD5		TWO-WAY DISCONNECTOR ON BUSBAR RISER																				
F	2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-34 20kV FlourMill		Scale				= H00		F				
	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant						Customer		PT WILMAR NABATI INDONESIA		+						
	0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.				Sh.	6							
	Ind.	Revision			Name	Approved	AI	Based on: BE318043			Customer Doc. Nbr.				Cont.	16							
1			2			3			4			5			6			7			8		

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1		2		3		4		5		6		7		8										
A	-TFA	ACTIVE POWER TRANSDUCERS						-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS						A									
	-TFC	CURRENT TRANSDUCERS						-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING															
	-TFF	FREQUENCY TRANSDUCERS						-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION															
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS															
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS															
	-TFM	MULTIFUNCTION TRANSDUCERS																						
B	-TFP	POWER-FACTOR TRANSDUCERS						-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS						B									
	-TFQ	REACTIVE POWER TRANSDUCERS						-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS															
	-TFS	SIGNAL CONVERTERS						-XD19	TERMINAL BLOCK FOR SPECIAL USE															
	-TFV	VOLTAGE TRANSDUCERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD															
	-WA	BUSBARS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS															
	-WBC	POWER CABLES						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT															
C	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDS	CURRENT SOCKETS						C									
	-WGC	CONTROL CABLES						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB															
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-WH	OPTICAL FIBRES						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK															
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS															
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1															
D	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2						D									
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3															
	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR															
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XDX	SUPPORT TERMINAL BLOCK															
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE															
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE															
E	-XDB10....-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE						E									
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE															
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE															
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE															
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT															
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																						
F	-XDC	CUSTOMER TERMINAL BLOCK													F									
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																						
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																						
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																						
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																						
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																						
F	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE													F									
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																						
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																						
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS																						
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																						
F	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS													F									
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS																						
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																						
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																						
2		Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-34 20kV FlourMill		Scale		= H00									
1		Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant						Customer	PT WILMAR NABATI INDONESIA	+									
0		First Issued	19.11.2018	YR	Checked	AW	Based on: BE318043		PT. ABB Sakti Industri				Drawing Nbr.		Sh. 7									
Ind.		Revision	Date	Name	Approved	AI							Customer Doc. Nbr.		Cont. 16									
1		2		3		4		5		6		7		8										

[illegible]

Panel	H01	H02	H03
Panel Name	SDC	SBC-W	SBC-W
Description	SDC - INCOMING FROM MVD-33	SBC-W - Incoming from MVD-11	SBC-W - Outgoing TR 34A Intake Silo 1600kVA
Rated Current	-	630A	630A
Max. Cable Size	-	2x300mm ²	2x300mm ²
Circuit-Diagram	312	304	306

No.	Revision	Date	Name	Approved	AI	Based on:	BE318043
2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant
1	Issued for Approval	18.12.2018	YR	Prepared	YR		
0	First Issued	19.11.2018	YR	Checked	AW		

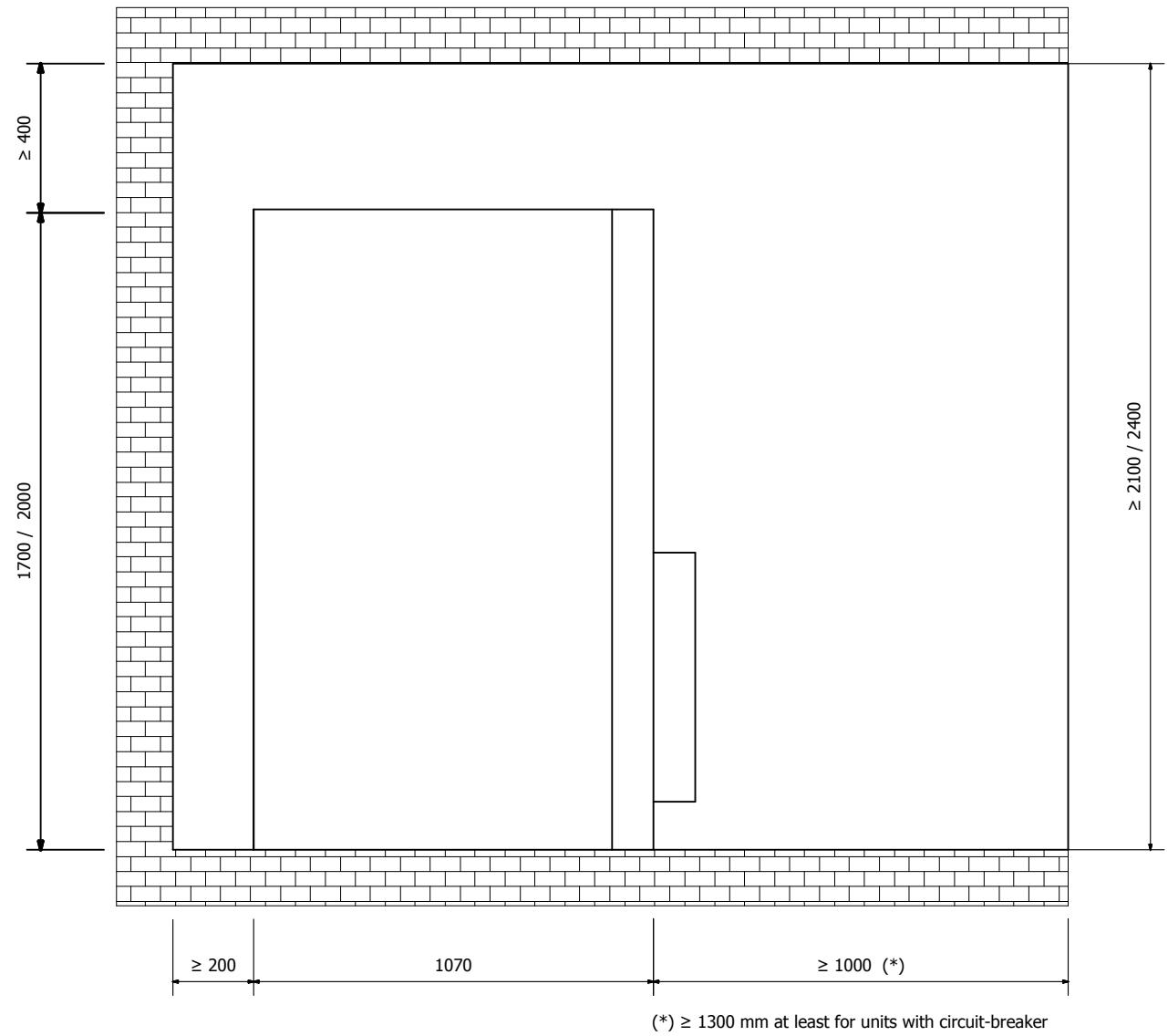
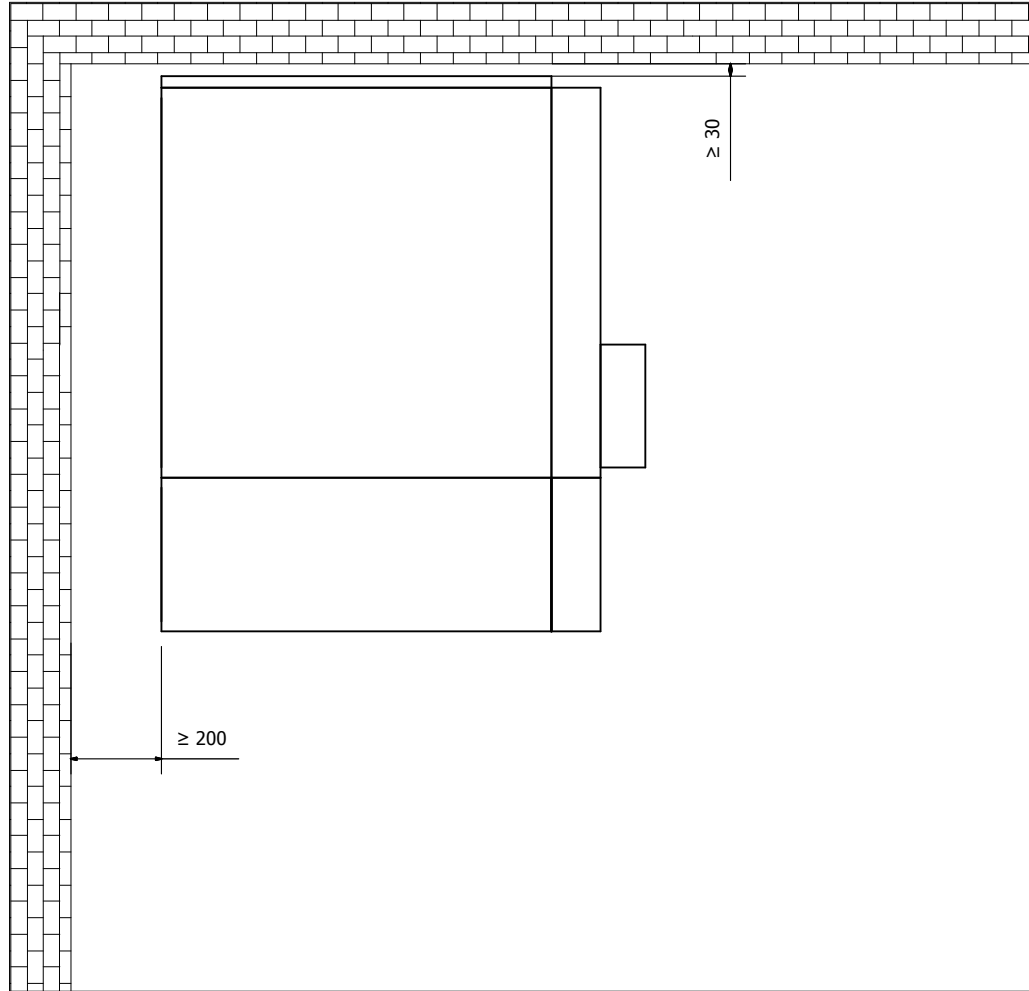
ABB EPMV PT. ABB Sakti Industri

SINGLE LINE DIAGRAM
SLD & GA Drawing
MVD-34 20kV FlourMill

Scale = H00
Customer PT WILMAR NABATI INDONESIA
Drawing Nbr. Sh. 8
Customer Doc. Nbr. Cont. 16

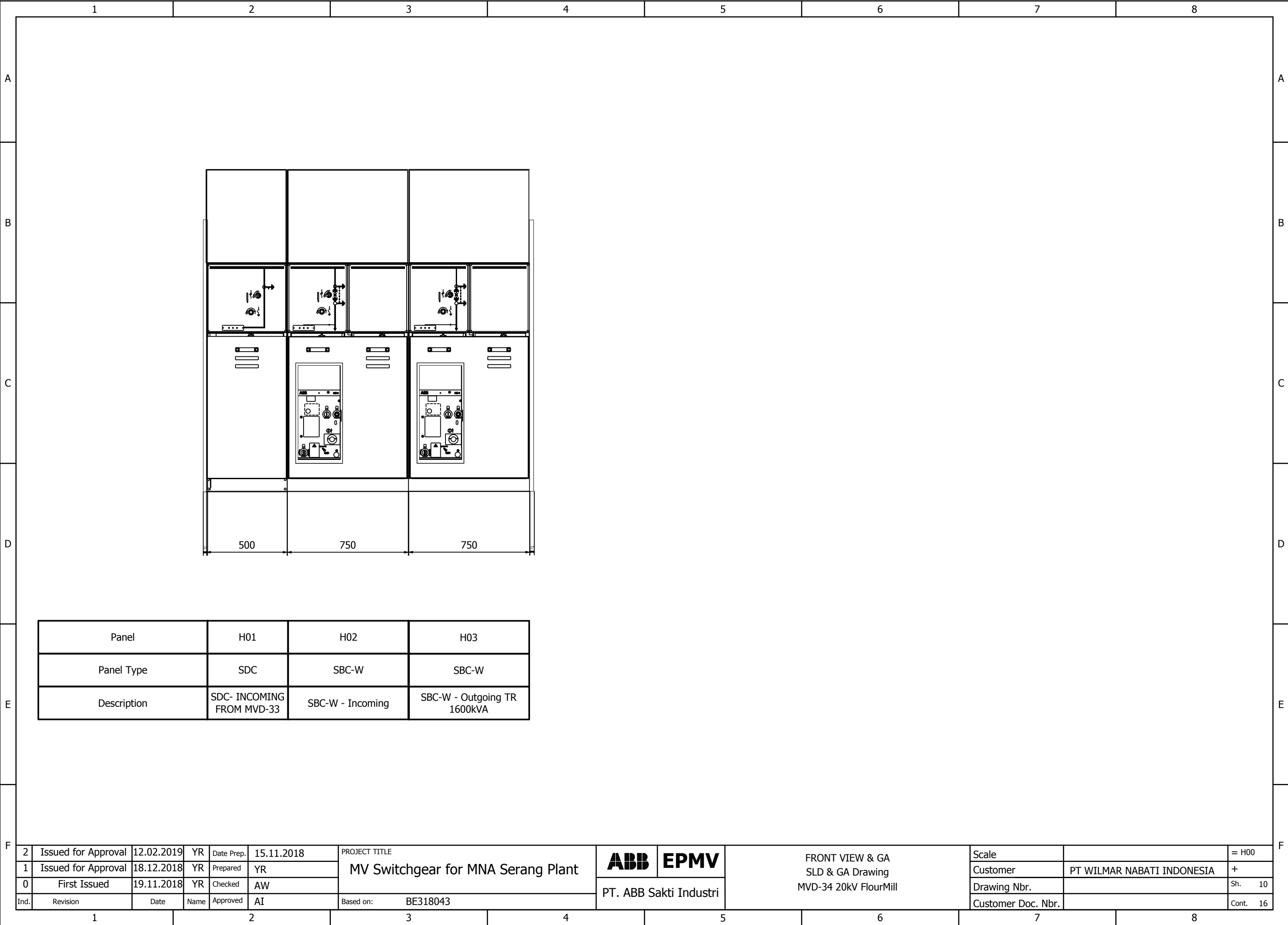
RULES FOR INSTALLATION IN THE ROOM

MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INSTALLATION RULES SLD & GA Drawing MVD-34 20kV FlourMill	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT WILMAR NABATI INDONESIA
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 9	
Ind.	Revision			Approved	AI		Based on: BE318043	Customer Doc. Nbr.				Cont. 16

[illegible]

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Panel	H01	H02	H03
Panel Type	SDC	SBC-W	SBC-W
Description	SDC- INCOMING FROM MVD-33	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA

2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-34 20kV FlourMill	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR				Customer	PT WILMAR NABATI INDONESIA	+
0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 10
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	Customer Doc. Nbr.		Cont.	16

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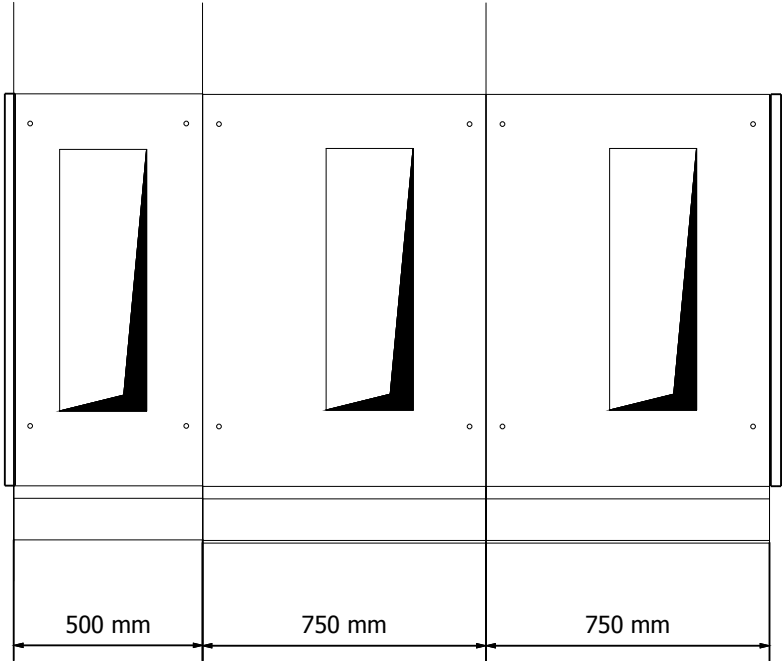
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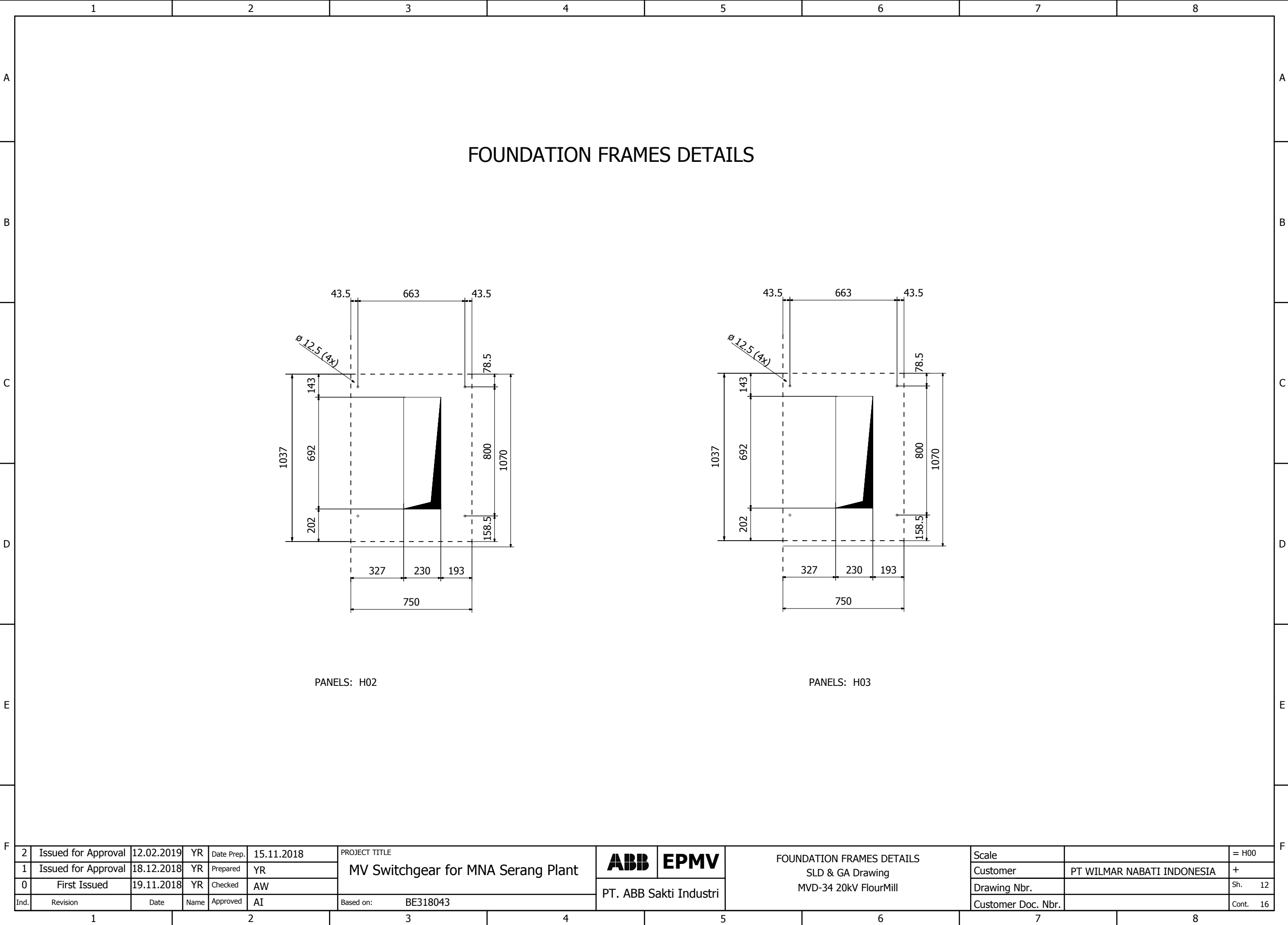
FOUNDATION FRAMES



Panel	H07	H08	H09
Panel Type	SDS	SBC-W	SBC-W
Description	SDS - Riser with VT	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA

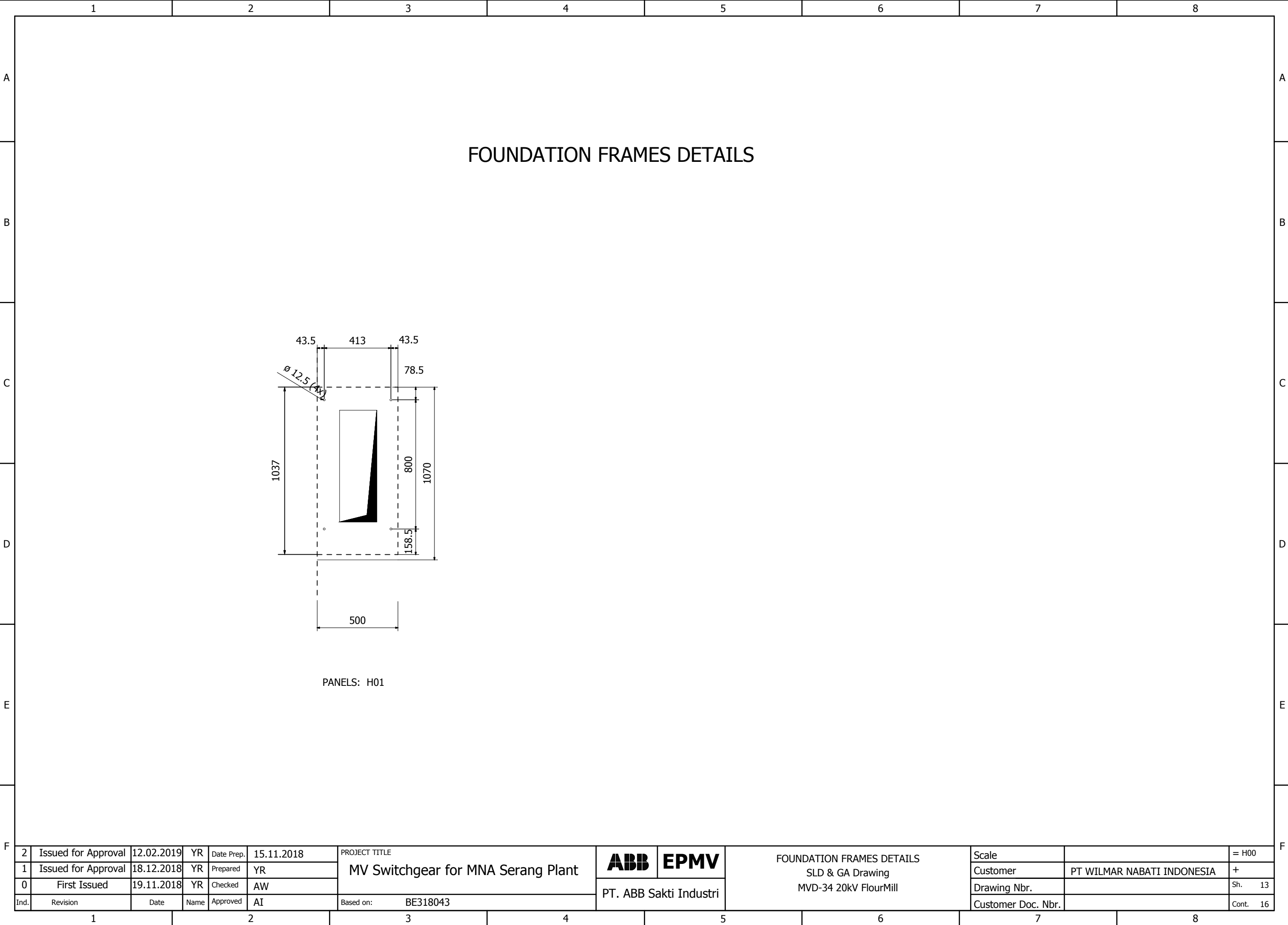
2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-34 20kV FlourMill	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	+
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-34 20kV FlourMill	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR				Customer	PT WILMAR NABATI INDONESIA	+
0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 12
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PT. ABB Sakti Industri		Customer Doc. Nbr.	Cont. 16

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1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	+
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 13		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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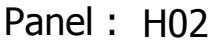
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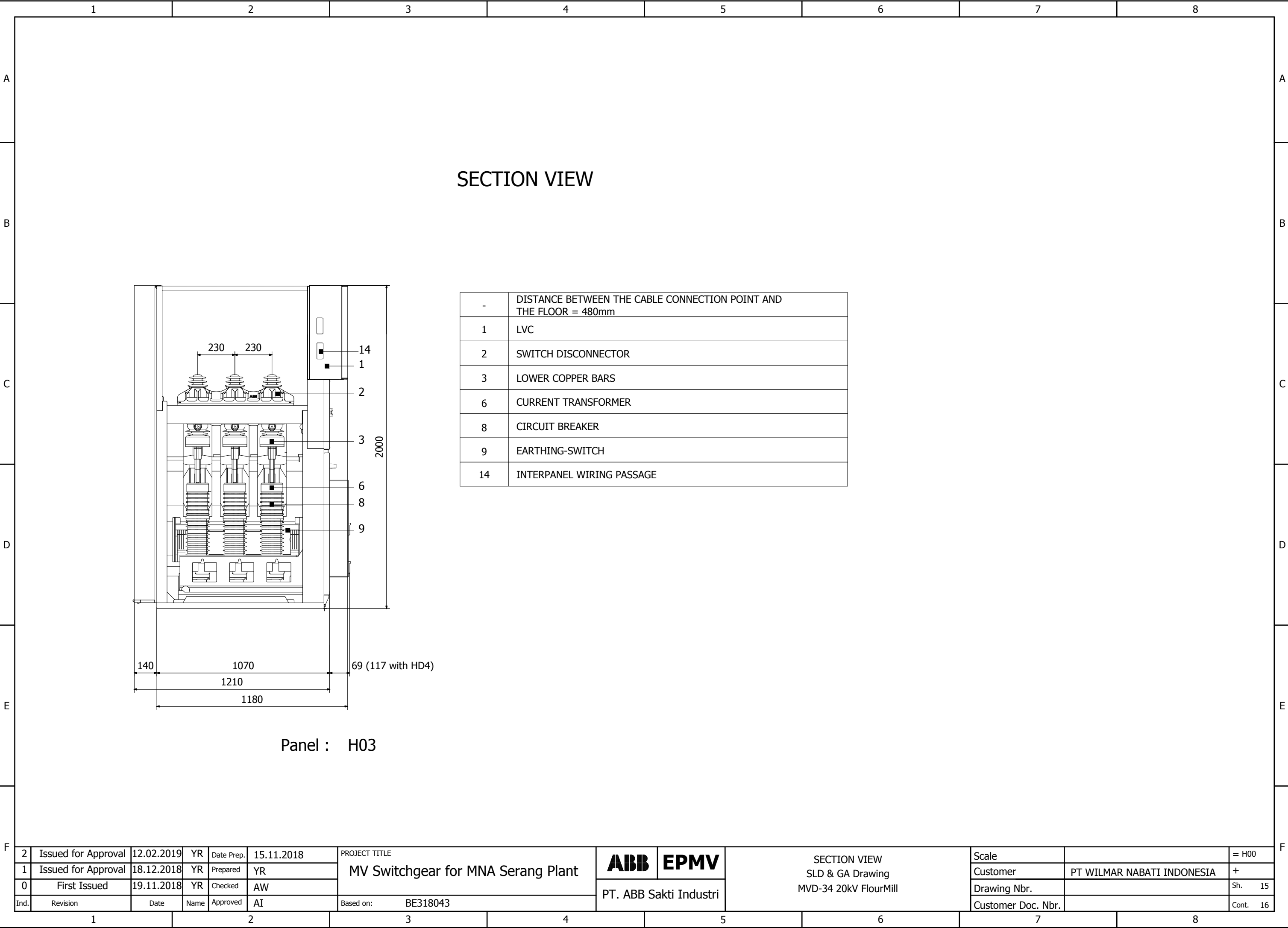
8

-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 480mm
1	LVC
2	SWITCH DISCONNECTOR
3	LOWER COPPER BARS
6	CURRENT TRANSFORMER
7	VOLTAGE TRANSFORMER
8	CIRCUIT BREAKER
9	EARTHING-SWITCH
14	INTERPANEL WIRING PASSAGE



2	Issued for Approval	12.02.2019	YR	Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-34 20kV FlourMill	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	+
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 14		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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